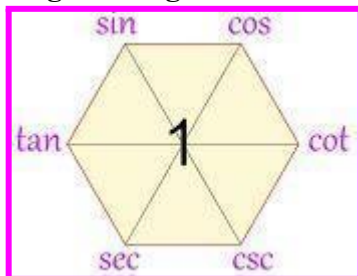


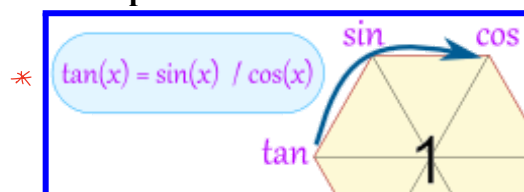
An **identity** is an equation that is true for all values of the variable(s). A **trigonometric identity** is an identity involving trigonometric functions.

Magic Hexagon



1. Quotient /Product Properties

a. Quotient Properties



Clockwise

$$\tan x = \frac{\sin x}{\cos x}, \quad \cot x = \frac{\csc x}{\sec x}$$

$$\sin x = \frac{\cos x}{\cot x}, \quad \csc x = \frac{\sec x}{\tan x}$$

$$\cos x = \frac{\cot x}{\csc x}, \quad \sec x = \frac{\tan x}{\sin x}$$

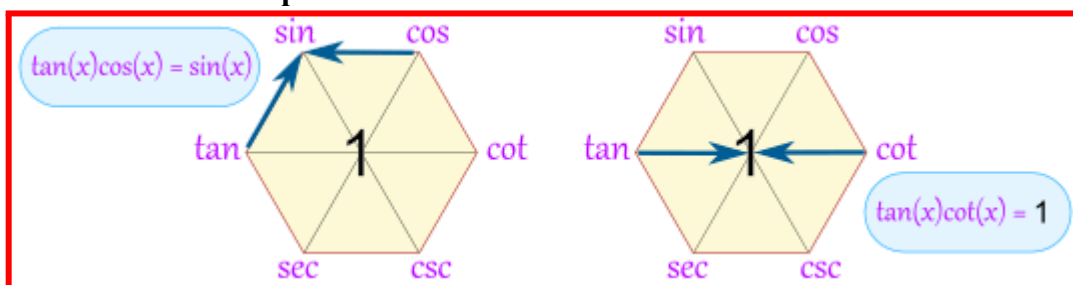
Counterclockwise

$$\tan x = \frac{\sec x}{\csc x}, \quad \cot x = \frac{\cos x}{\sin x}$$

$$\sec x = \frac{\csc x}{\cot x}, \quad \cos x = \frac{\sin x}{\tan x}$$

$$\csc x = \frac{\cot x}{\cos x}, \quad \sin x = \frac{\tan x}{\sec x}$$

b. Product Properties



$$\tan x \cos x = \sin x$$

$$\sin x \cot x = \cos x$$

$$\cos x \csc x = \cot x$$

$$\cot x \sec x = \csc x$$

$$\csc x \tan x = \sec x$$

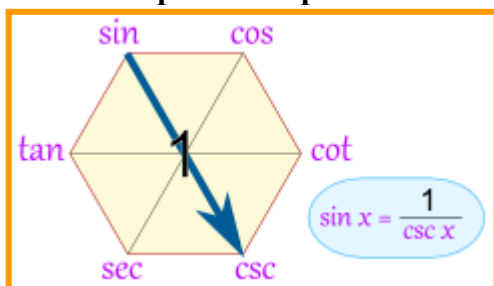
$$\sec x \sin x = \tan x$$

$$\tan x \cot x = 1$$

$$\sec x \cos x = 1$$

$$\sin x \csc x = 1$$

2. Reciprocal Properties

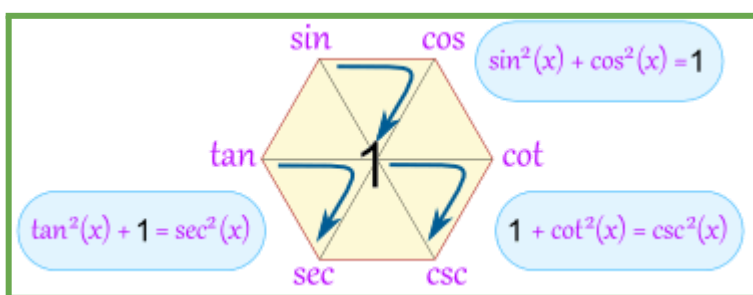


$$\sin x = \frac{1}{\csc x}, \quad \csc x = \frac{1}{\sin x}$$

$$\cos x = \frac{1}{\sec x}, \quad \sec x = \frac{1}{\cos x}$$

$$\tan x = \frac{1}{\cot x}, \quad \cot x = \frac{1}{\tan x}$$

3. Pythagorean Properties



$$\sin^2 x + \cos^2 x = 1$$

$$\sin^2 x = 1 - \cos^2 x$$

$$\cos^2 x = 1 - \sin^2 x$$

$$1 + \cot^2 x = \csc^2 x$$

$$\cot^2 x = \csc^2 x - 1$$

$$1 = \csc^2 x - \cot^2 x$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\tan^2 x = \sec^2 x - 1$$

$$1 = \sec^2 x - \tan^2 x$$

List even-odd identities for trigonometric functions.

Even

$$\cos x, \cos(-x) = \cos x$$

$$\sec x, \sec(-x) = \sec x$$

Odd

$$\sin x, \sin(-x) = -\sin x$$

$$\tan x, \tan(-x) = -\tan x$$

$$\csc x, \csc(-x) = -\csc x$$

$$\cot x, \cot(-x) = -\cot x$$

List the six trigonometric **COFUNCTION** identities.

Memorize ✱

$\sin\left(\frac{\pi}{2} - x\right) = \cos x$	$\tan\left(\frac{\pi}{2} - x\right) = \cot x$	$\sec\left(\frac{\pi}{2} - x\right) = \csc x$
$\cos\left(\frac{\pi}{2} - x\right) = \sin x$	$\cot\left(\frac{\pi}{2} - x\right) = \tan x$	$\csc\left(\frac{\pi}{2} - x\right) = \sec x$

I. Simplifying Trigonometric Expressions

Identities enable us to write the same expression in different ways. It is often possible to rewrite a complicated-looking expression as a much simpler one. To simplify trigonometric expressions, we use factoring, common denominators, Special Product Formulas, and the fundamental trigonometric identities.

II. Proving Trigonometric Identities

How can you tell if a given equation is not an identity?

It's easy to decide when a given equation is not an identity. All we need to do is show that the equation does not hold for some value of the variable or variables.

Guidelines for proving trigonometric identities.

1. **Start with one side.** Pick one side of the equation and write it down. Your goal is to transform it into the other side. It's usually easier to start with the more complicated side.
2. **Use known identities.** Use algebra and the identities you know to change the side you started with. Bring fractional expressions to a common denominator, factor, and use the fundamental identities to simplify expressions.
3. **Convert to sines and cosines.** If you are stuck, you may find it helpful to rewrite all functions in terms of sines and cosines.

Here is another method for proving that an equation is an identity: if we can transform each side of the equation *separately*, by way of identities, to arrive at the same result, then the equation is an identity.

Example 1: Write the trigonometric expression in terms of sine and cosine, and then simplify.

a. $\sin \theta \sec \theta$

$$= \sin \theta \cdot \frac{1}{\cos \theta}$$

$$= \frac{\sin \theta}{\cos \theta}$$

c. $\sin^2\left(\frac{\pi}{2} - y\right) \sec y$

$$= \left[\sin\left(\frac{\pi}{2} - y\right)\right]^2 \sec y$$

$$= (\cos y)^2 \cdot \frac{1}{\cos y}$$

$$= \cos y$$

b. $\tan^2 x - \sec^2 x$

$$= \frac{\sin^2 x}{\cos^2 x} - \frac{1}{\cos^2 x}$$

$$= \frac{\sin^2 x - 1}{\cos^2 x}$$

$$= \frac{-1(1 - \sin^2 x)}{\cos^2 x}$$

$$= -\frac{\cos^2 x}{\cos^2 x} = -1$$

d. $\frac{\cot t}{\csc t - \sin t}$

$$= \frac{\frac{\cos t}{\sin t}}{\frac{1}{\sin t} - \sin t}$$

$$= \frac{\cos t}{\sin t} \cdot \frac{\sin t}{1 - \sin^2 t}$$

$$= \frac{\cos t \cancel{\sin t}}{\cancel{\sin t} (1 - \sin^2 t)}$$

$$= \frac{\cos t}{\cos^2 t}$$

$$= \frac{1}{\cos t} = \sec t$$

Example 2: Simplify the trigonometric expression.

a. $\frac{\cos x \sec x}{\cot x}$

$$= \frac{1}{\cot x}$$

$$= \tan x$$

b. $\frac{\cos(-x)}{\sec(-x) + \tan x}$

$$= \frac{\cos x}{\sec x + \tan x}$$

$$= \frac{\cos x}{\frac{1}{\cos x} + \frac{\sin x}{\cos x}}$$

$$= \frac{\cos x}{\frac{1 + \sin x}{\cos x}}$$

$$= \frac{\cos^2 x}{1 + \sin x}$$

c. $(\sin^4 \alpha - \cos^4 \alpha) + \cos^2 \alpha$

$$= (\sin^2 \alpha - \cos^2 \alpha)(\sin^2 \alpha + \cos^2 \alpha) + \cos^2 \alpha$$

$$= \sin^2 \alpha - \cancel{\cos^2 \alpha} + \cancel{\cos^2 \alpha}$$

$$= \sin^2 \alpha$$

factor using difference of 2 squares

$$\begin{aligned}
 \text{d. } & \frac{1 + \cos(y - \frac{\pi}{2})}{1 + \csc y} \\
 &= \frac{1 + \cos(-(\frac{\pi}{2} - y))}{1 + \csc y} \\
 &= \frac{1 + \cos(\frac{\pi}{2} - y)}{1 + \csc y} \\
 &= \frac{1 + \sin y}{1 + \frac{1}{\sin y}} \\
 &= \frac{1 + \sin y}{\frac{\sin y + 1}{\sin y}} \\
 &= \cancel{1 + \sin y} \cdot \frac{\sin y}{\cancel{1 + \sin y}} \\
 &= \boxed{\sin y}
 \end{aligned}$$

$$\begin{aligned}
 \text{e. } & \frac{1 + \sin u}{\cos u} + \frac{\cos u}{1 + \sin u} \\
 &= \frac{(1 + \sin u)^2 + \cos^2 u}{(\cos u)(1 + \sin u)} \\
 &= \frac{1 + 2\sin u + \sin^2 u + \cos^2 u}{(\cos u)(1 + \sin u)} \\
 &= \frac{2 + 2\sin u}{(\cos u)(1 + \sin u)} \\
 &= \frac{2(1 + \sin u)}{(\cos u)(1 + \sin u)} \\
 &= \frac{2}{\cos u} \\
 &= \boxed{2 \sec u}
 \end{aligned}$$

$$\begin{aligned}
 \text{f. } & \frac{2 + \tan^2 x}{\sec^2 x} - 1 \\
 &= \frac{2 + \tan^2 x - \sec^2 x}{\sec^2 x} \\
 &= \frac{2 + (- (\sec^2 x - \tan^2 x))}{\sec^2 x} \\
 &= \frac{2 - 1}{\sec^2 x} \\
 &= \frac{1}{\sec^2 x} \\
 &= \boxed{\cos^2 x}
 \end{aligned}$$

Example 3: Verify the identity using the fundamental trigonometric identities.

$$\text{a. } \frac{\cot x \sec x}{\csc x} = 1$$

$$\begin{aligned}
 \text{LHS: } & \frac{\cot x \sec x}{\csc x} \\
 &= \frac{\frac{\cos x}{\sin x} \cdot \frac{1}{\cos x}}{\frac{1}{\sin x}} \\
 &= 1 \\
 &= \text{RHS}
 \end{aligned}$$

$$\text{b. } \tan\left(x - \frac{\pi}{2}\right) \sin x = \sin\left(x - \frac{\pi}{2}\right)$$

$$\begin{aligned}
 \text{LHS: } & \tan\left(-\left(\frac{\pi}{2} - x\right)\right) \sin x \\
 &= -\tan\left(\frac{\pi}{2} - x\right) \sin x \\
 &= -\cot x \sin x \\
 &= -\cos x \\
 &= -\sin\left(\frac{\pi}{2} - x\right) \\
 &= \sin\left(x - \frac{\pi}{2}\right) = \text{RHS}
 \end{aligned}$$

$$\text{c. } \cos(-x) - \sin(-x) = \cos x + \sin x$$

$$\begin{aligned}
 \text{LHS: } & \cos(-x) - \sin(-x) \\
 &= \cos x - (-\sin x) \\
 &= \cos x + \sin x \\
 &= \text{RHS}
 \end{aligned}$$

$$\text{d. } \frac{\sec A - 1}{\sec A + 1} = \frac{1 - \cos A}{1 + \cos A}$$

$$\begin{aligned}
 \text{LHS: } & \frac{\frac{1}{\cos A} - 1}{\frac{1}{\cos A} + 1} \\
 &= \frac{\frac{1 - \cos A}{\cos A}}{\frac{1 + \cos A}{\cos A}} \\
 &= \frac{1 - \cos A}{1 + \cos A} \\
 &= \text{RHS}
 \end{aligned}$$

$$\text{e. } \csc x - \sin x = \cos x \cot x$$

$$\begin{aligned}
 \text{LHS: } & \csc x - \sin x \\
 &= \frac{1}{\sin x} - \sin x \\
 &= \frac{1 - \sin^2 x}{\sin x} \\
 &= \frac{\cos^2 x}{\sin x} \\
 &= \cos x \cdot \frac{\cos x}{\sin x} \\
 &= \cos x \cot x \\
 &= \text{RHS}
 \end{aligned}$$

$$\text{f. } \frac{1 + \sin x}{1 - \sin x} = (\tan x + \sec x)^2$$

$$\begin{aligned}
 \text{RHS: } & \left(\frac{\sin x}{\cos x} + \frac{1}{\cos x}\right)^2 \\
 &= \left(\frac{\sin x + 1}{\cos x}\right)^2 \\
 &= \frac{(1 + \sin x)^2}{\cos^2 x} \\
 &= \frac{(1 + \sin x)^2}{1 - \sin^2 x} \\
 &= \frac{(1 + \sin x)(1 + \sin x)}{(1 + \sin x)(1 - \sin x)} \\
 &= \frac{(1 + \sin x)}{(1 - \sin x)} \\
 &= \text{LHS}
 \end{aligned}$$

$$\text{g. } \sin^2 y + \cos^2 y + \tan^2 y = \sec^2 y$$

$$\begin{aligned} \text{LHS: } & \sin^2 y + \cos^2 y + \tan^2 y \\ &= 1 + \tan^2 y \\ &= \sec^2 y \\ &= \text{RHS} \end{aligned}$$

$$\text{h. } \frac{2 \sin x \cos x}{(\sin x + \cos x)^2 - 1} = 1$$

$$\begin{aligned} \text{LHS: } & \frac{2 \sin x \cos x}{\sin^2 x + 2 \sin x \cos x + \cos^2 x - 1} \\ &= \frac{2 \sin x \cos x}{2 \sin x \cos x} \\ &= \frac{2 \sin x \cos x}{2 \sin x \cos x} \\ &= 1 \\ &= \text{RHS} \end{aligned}$$

$$\text{i. } \sin^4 \theta - \cos^4 \theta = \sin^2 \theta - \cos^2 \theta$$

$$\begin{aligned} \text{LHS: } & \sin^4 \theta - \cos^4 \theta \\ &= (\sin^2 \theta + \cos^2 \theta)(\sin^2 \theta - \cos^2 \theta) \\ &= 1(\sin^2 \theta - \cos^2 \theta) \\ &= \sin^2 \theta - \cos^2 \theta \\ &= \text{RHS} \end{aligned}$$

$$\text{j. } \frac{1 + \tan^2 u}{1 - \tan^2 u} = \frac{1}{\cos^2 u - \sin^2 u}$$

$$\begin{aligned} \text{LHS: } & \frac{1 + \frac{\sin^2 u}{\cos^2 u}}{1 - \frac{\sin^2 u}{\cos^2 u}} \\ &= \frac{\frac{\cos^2 u + \sin^2 u}{\cos^2 u}}{\frac{\cos^2 u - \sin^2 u}{\cos^2 u}} \\ &= \frac{1}{\cos^2 u - \sin^2 u} \end{aligned}$$

$$\text{k. } \frac{\cos \theta}{1 - \sin \theta} = \frac{\sin \theta - \csc \theta}{\cos \theta - \cot \theta}$$

$$\begin{aligned} \text{RHS: } & \frac{\sin \theta - \frac{1}{\sin \theta}}{\cos \theta - \frac{\cos \theta}{\sin \theta}} \\ &= \frac{\frac{\sin^2 \theta - 1}{\sin \theta}}{\frac{\cos \theta \sin \theta - \cos \theta}{\sin \theta}} \\ &= \frac{\sin^2 \theta - 1}{\cos \theta \sin \theta - \cos \theta} \\ &= \frac{(\sin \theta - 1)(\sin \theta + 1)}{\cos \theta (\sin \theta - 1)} \\ &= \frac{\sin \theta + 1}{\cos \theta} \\ &= \frac{(\sin \theta + 1)(1 - \sin \theta)}{\cos \theta \cdot (1 - \sin \theta)} \\ &= \frac{1 - \sin^2 \theta}{\cos \theta (1 - \sin \theta)} \\ &= \frac{\cos^2 \theta}{\cos \theta (1 - \sin \theta)} \\ &= \frac{\cos \theta}{1 - \sin \theta} \\ &= \text{LHS} \end{aligned}$$

$$\text{l. } \frac{\tan x + \tan y}{\cot x + \cot y} = \tan x \tan y$$

$$\begin{aligned} \text{LHS: } & \frac{\tan x + \tan y}{\cot x + \cot y} \\ &= \frac{\tan x + \tan y}{\frac{1}{\tan x} + \frac{1}{\tan y}} \\ &= \frac{\tan x + \tan y}{\frac{\tan y + \tan x}{\tan x \tan y}} \\ &= \tan x \tan y \\ &= \text{RHS} \end{aligned}$$

Memorize
★

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

m. $\frac{\sin^3 y - \csc^3 y}{\sin y - \csc y} = \sin^2 y + \csc^2 y + 1$

LHS: $\frac{(\sin y - \csc y)(\sin^2 y + \sin y \csc y + \csc^2 y)}{\sin y - \csc y}$

$$= \sin^2 y + \sin y \csc y + \csc^2 y$$

$$= \sin^2 y + \csc^2 y + 1$$

$$= \text{RHS}$$

n. $\frac{\sin w}{\sin w + \cos w} = \frac{\tan w}{1 + \tan w}$

LHS: $\frac{\sin w}{\sin w + \cos w}$

$$= \frac{\sin w}{\sin w + \cos w} \cdot \frac{\sec w}{\sec w}$$

$$= \frac{\sin w \sec w}{\sin w \sec w + \cos w \sec w}$$

$$= \frac{\tan w}{\tan w + 1}$$

$$= \text{RHS}$$

o. $\sec v - \tan v = \frac{1}{\sec v + \tan v}$

RHS: $\frac{1}{\sec v + \tan v} \cdot \frac{\sec v - \tan v}{\sec v - \tan v}$

$$= \frac{\sec v - \tan v}{\sec^2 v - \tan^2 v}$$

$$= \frac{\sec v - \tan v}{1}$$

$$= \sec v - \tan v$$

$$= \text{LHS}$$

p. $\frac{\sec u - 1}{\sec u + 1} = \frac{\tan u - \sin u}{\tan u + \sin u}$

RHS: $\frac{\frac{\sin u}{\cos u} - \sin u}{\frac{\sin u}{\cos u} + \sin u}$

$$= \frac{\frac{\sin u - \sin u \cos u}{\cos u}}{\frac{\sin u + \sin u \cos u}{\cos u}}$$

$$= \frac{\sin u - \sin u \cos u}{\sin u + \sin u \cos u}$$

$$= \frac{\sin u (1 - \cos u)}{\sin u (1 + \cos u)}$$

$$= \frac{1 - \cos u}{1 + \cos u}$$

$$= \frac{\frac{1}{\cos u} - 1}{\frac{1}{\cos u} + 1}$$

$$= \frac{\sec u - 1}{\sec u + 1}$$

$$= \text{LHS}$$

Example 4: Make the indicated trigonometric substitution in the given algebraic expression and simplify. Assume that $0 < \theta < \frac{\pi}{2}$.

a. $\sqrt{x^2 - 1}, x = \sec \theta$

$$= \sqrt{\sec^2 \theta - 1}$$

$$= \sqrt{\tan^2 \theta}$$

$$= \boxed{\tan \theta}$$

b. $\frac{1}{x^2 \sqrt{4+x^2}}, x = 2 \tan \theta$

$$= \frac{1}{4 \tan^2 \theta \sqrt{4 + 4 \tan^2 \theta}}$$

$$= \frac{1}{4 \tan^2 \theta \sqrt{4(1 + \tan^2 \theta)}}$$

$$= \frac{1}{8 \tan^2 \theta \sqrt{\sec^2 \theta}}$$

$$= \frac{1}{8} \frac{1}{\tan^2 \theta} \frac{1}{\sec \theta}$$

$$= \frac{1}{8} \cot^2 \theta \cos \theta$$

c. $\frac{\sqrt{x^2 - 25}}{x}, x = 5 \sec \theta$

$$= \frac{\sqrt{25 \sec^2 \theta - 25}}{5 \sec \theta}$$

$$= \frac{\sqrt{25(\sec^2 \theta - 1)}}{5 \sec \theta}$$

$$= \frac{5 \tan \theta}{5 \sec \theta}$$

$$= \boxed{\sin \theta}$$